

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical - IV

Roll No.: T032	Name: KHUSHI
Class: TYIT	Batch: 2
Date of Assignment: 20-8-26	Date/Time of Submission: 20-8-26

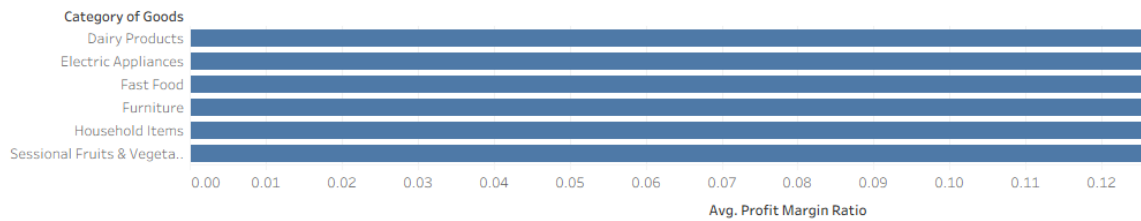
Performing Data Analysis using Calculations in Tableau :

a) Create basic calculated fields using formulas

The screenshot illustrates the steps to create a calculated field in Tableau:

- Field List:** Shows 'Longitude (generated)', 'store_sales_data (2)-sele...', and 'Measure Values'.
- Right-click Menu:** A context menu is open over the 'Data Source' pane, with 'Create Calculated Field...' selected.
- Calculated Field Dialog:** A dialog box titled 'Profit Margin Ratio' shows the formula $[Profit] / [Sales]$. It includes an 'Apply' button and an 'OK' button. A message at the bottom states 'The calculation is valid.'
- Visualization:** A horizontal bar chart is shown with 'Category of Goods' on the x-axis and 'SUM(Profit Margin Ratio)' on the y-axis. The bars represent different goods categories.
- Field Properties Menu:** A context menu is open over the 'SUM(Profit Margin Ratio)' field, showing options like 'Filter...', 'Show Filter', 'Format...', 'Show Header', 'Include in Tooltip', 'Find in Data pane', 'Dimension', 'Attribute', 'Measure (Sum)', 'Discrete', 'Continuous', 'Edit in Shelf', 'Add Table Calculation...', 'Quick Table Calculation', and 'Remove'.

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b) Perform row-level calculations

store_sales_data (2)-sel...

Search   

Create Calculated Field...

Create Parameter...

Group by Folder

- Group by Data Source Table
- Sort by Name
- Sort by Data Source Order

Hide All Unused Fields

Show Hidden Fields

Expand All

Collapse All

Filters

Marks

Automatic

Color

Size

Detail

Tooltip

Abc Product ID

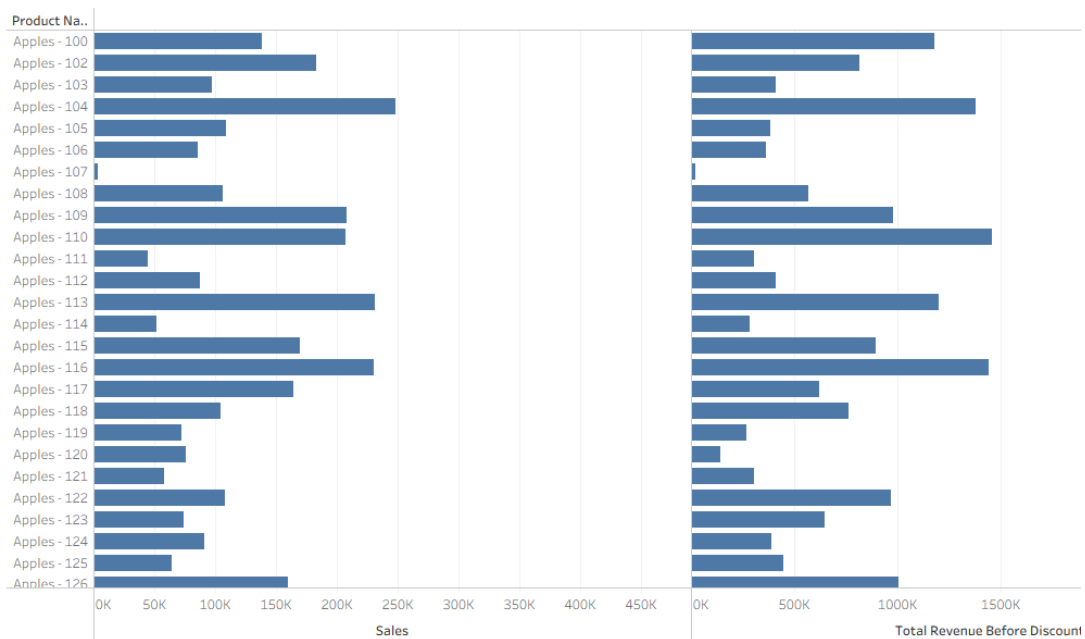
Total Revenue Before Discount

$[Sales] * [Quantity]$

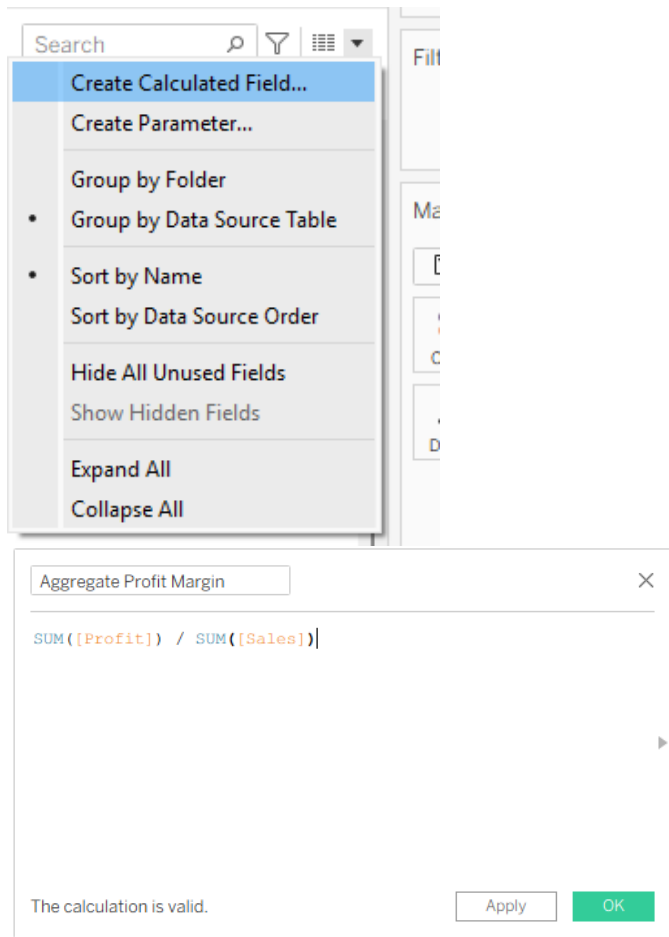
The calculation is valid.

Apply OK

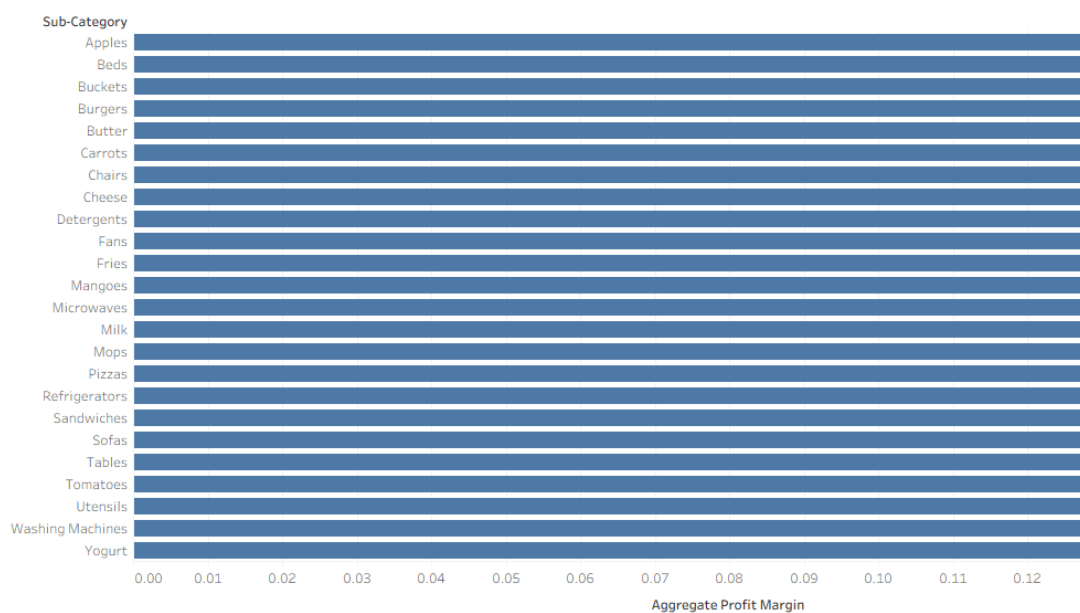
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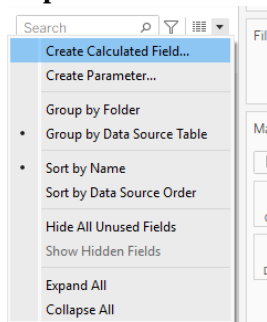
c) Perform aggregate-level calculations



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d) Implement FIXED Level of Detail (LOD) calculations



Fixed Regional Sales

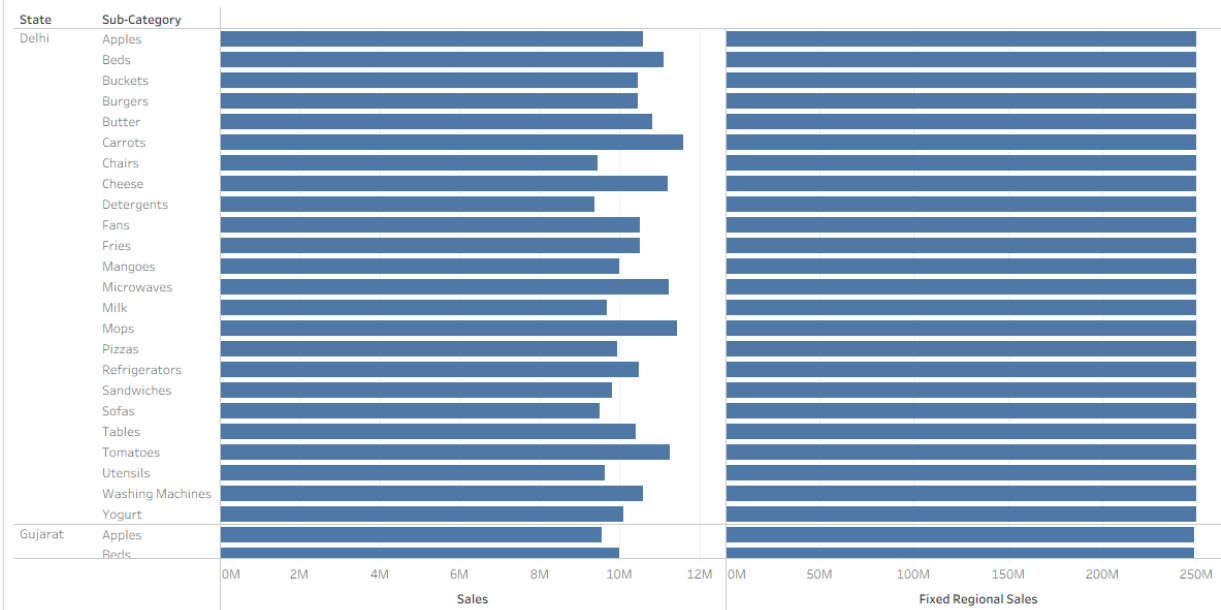


```
{ FIXED [State] : SUM([Sales]) }
```

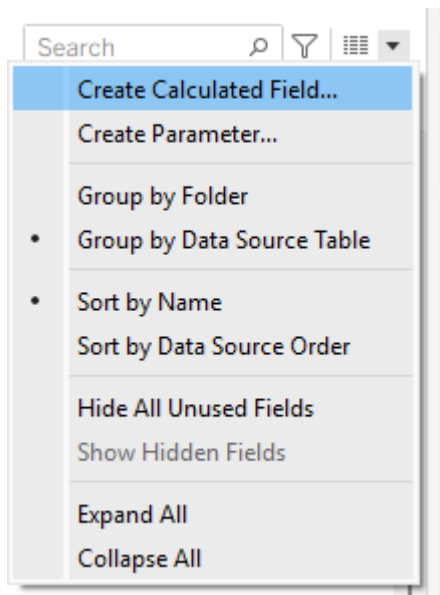
The calculation is valid.

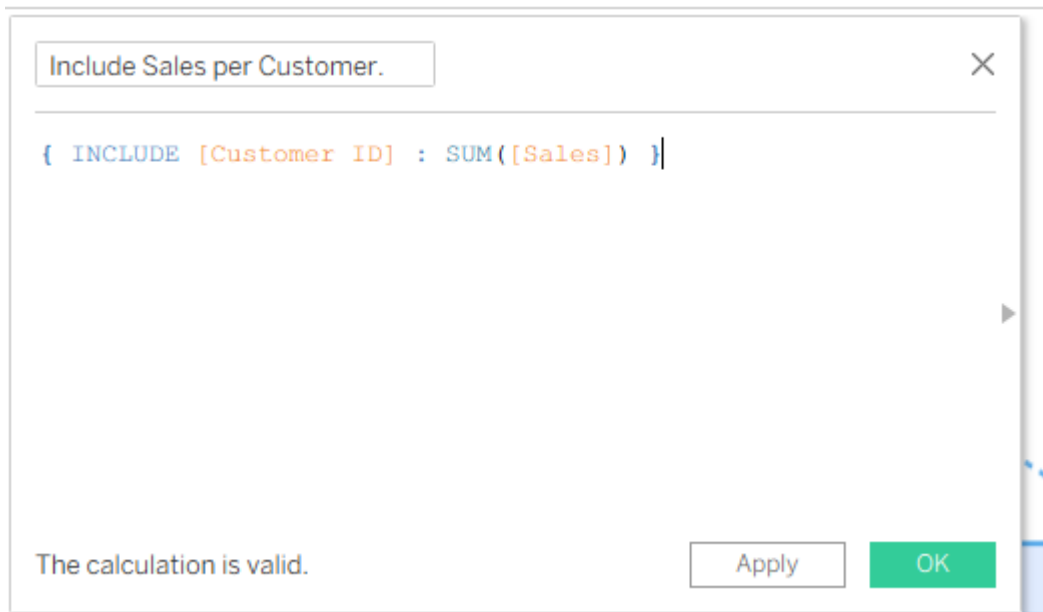
Apply

OK

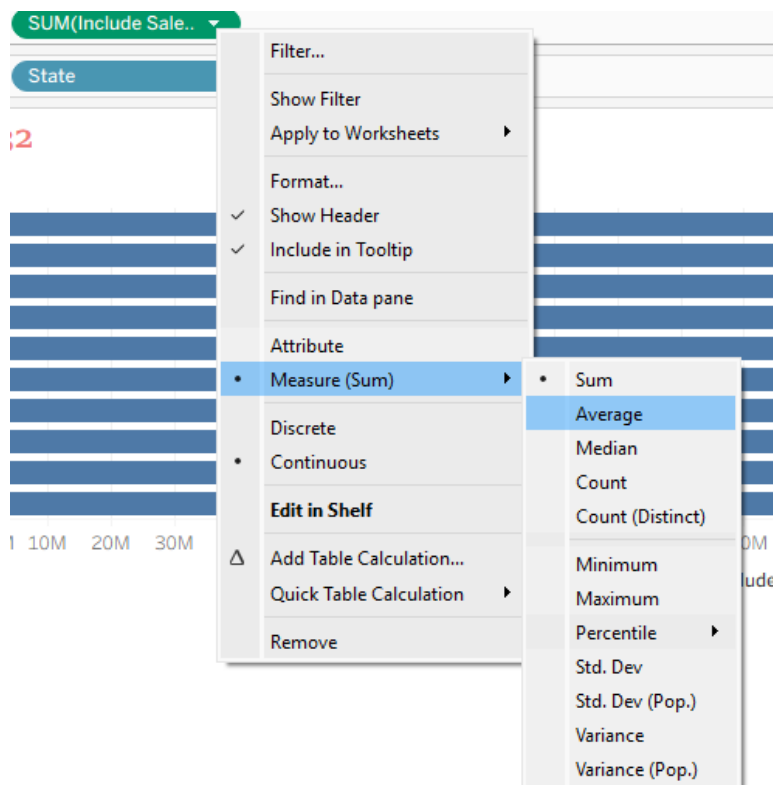
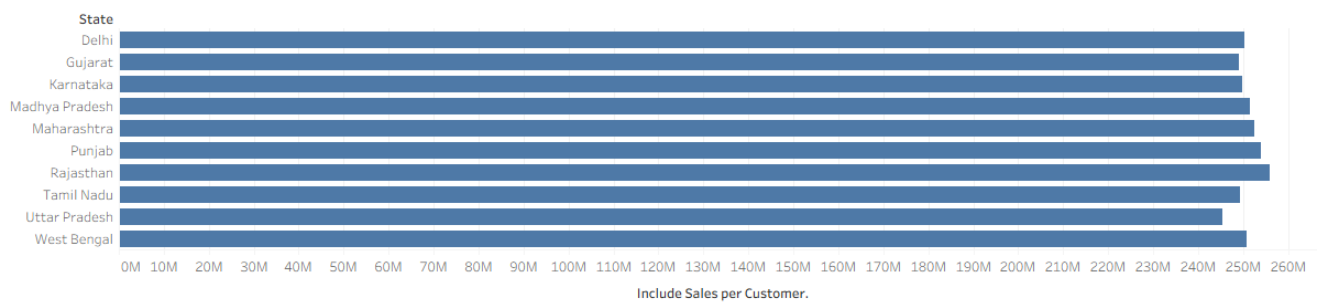
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- e) Use INCLUDE and EXCLUDE LOD expressions
INCLUDE :

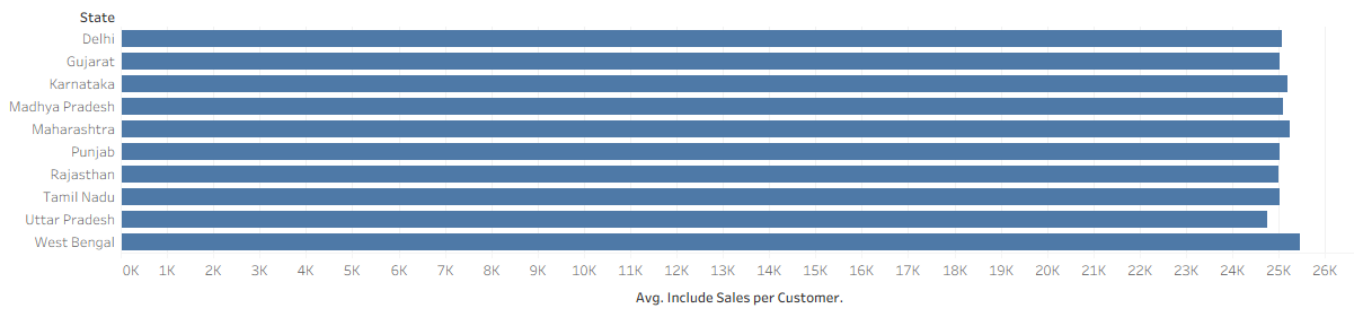




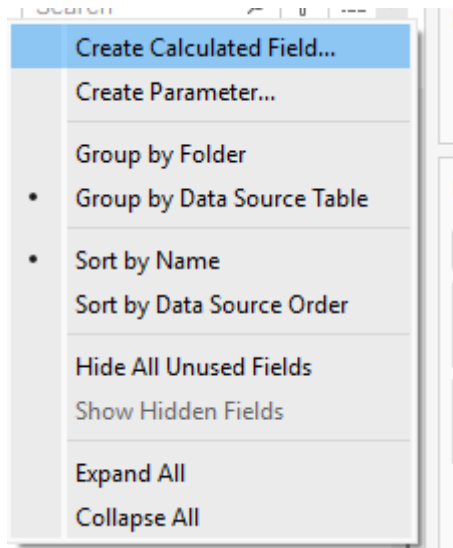
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EXCLUDE :



Exclude Category Sales

×

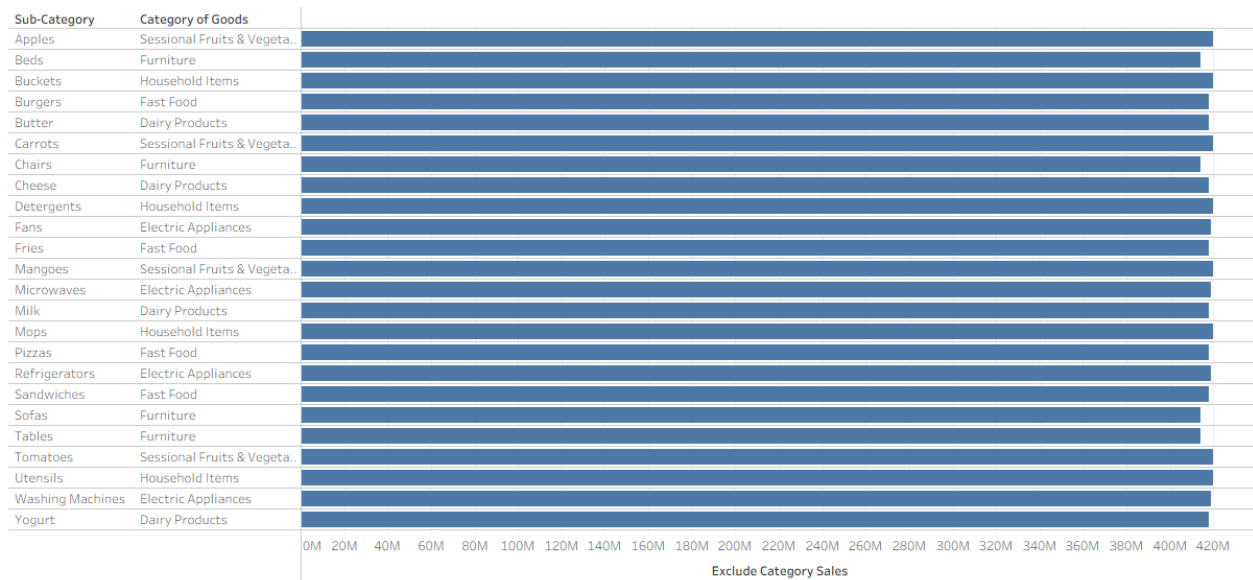
```
{ EXCLUDE [Sub-Category] : SUM([Sales]) }
```

The calculation is valid.

Apply

OK

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f) Create and use parameters for dynamic analysis

Search

- Create Calculated Field...
- Create Parameter...
- Group by Folder
 - Group by Data Source Table
- Sort by Name
 - Sort by Data Source Order
- Hide All Unused Fields
- Show Hidden Fields
- Expand All
- Collapse All

Create Parameter

Name
Select Profit Threshold

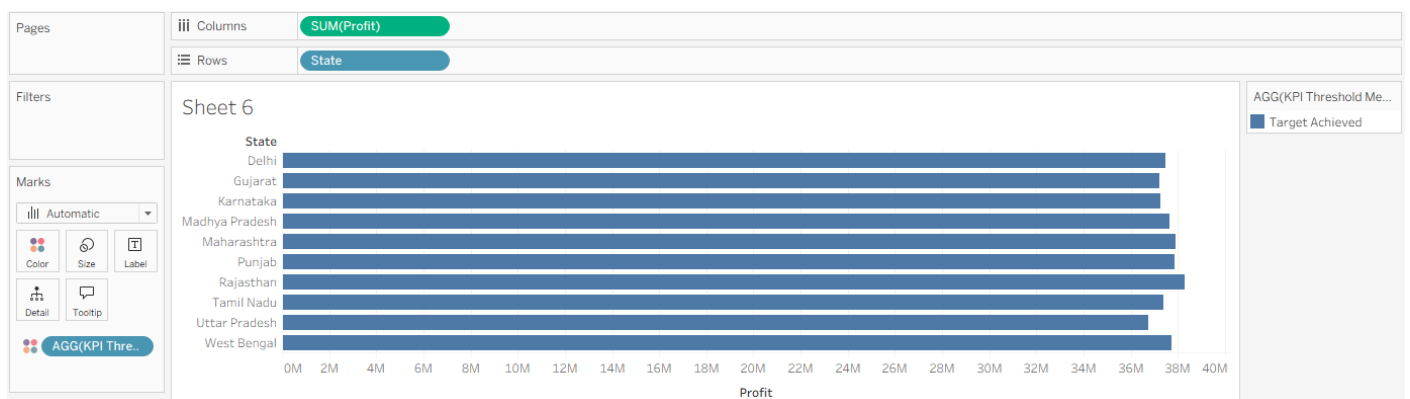
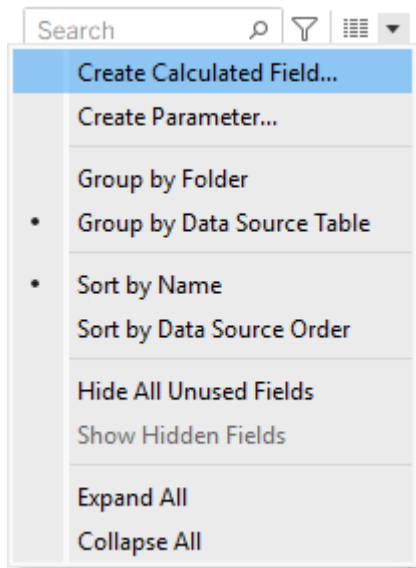
Properties

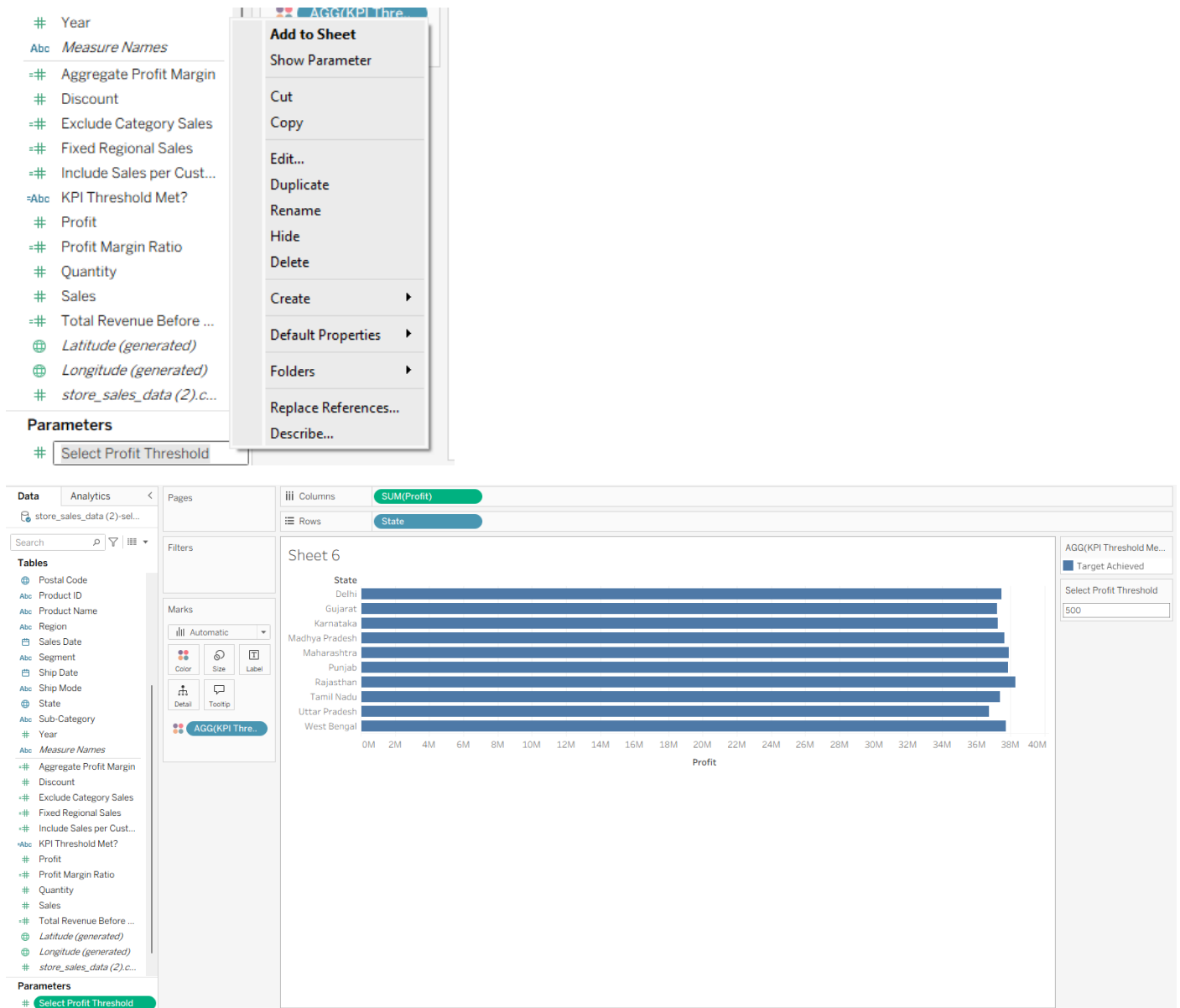
Data type: Float Display format: 500

Current value: 500 Value when workbook opens: Current value

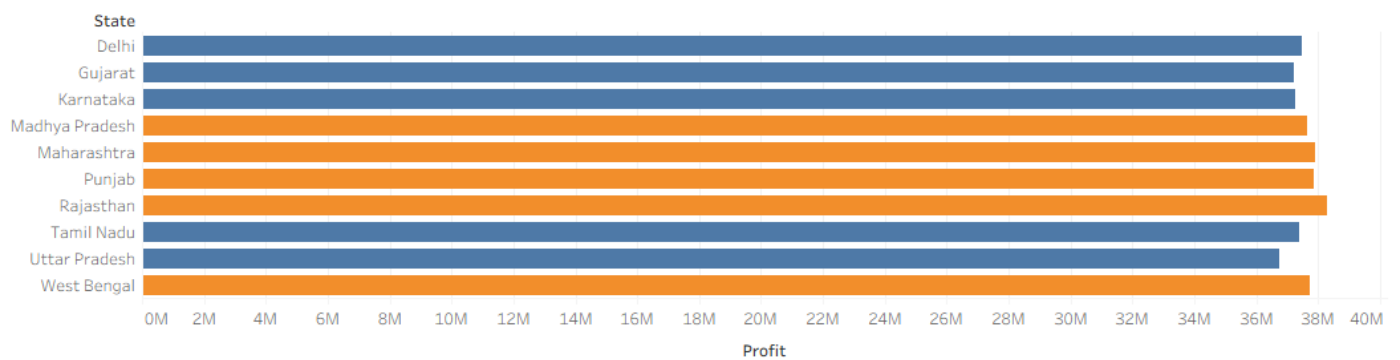
Allowable values
☒ All ☐ List ☐ Range

Cancel OK

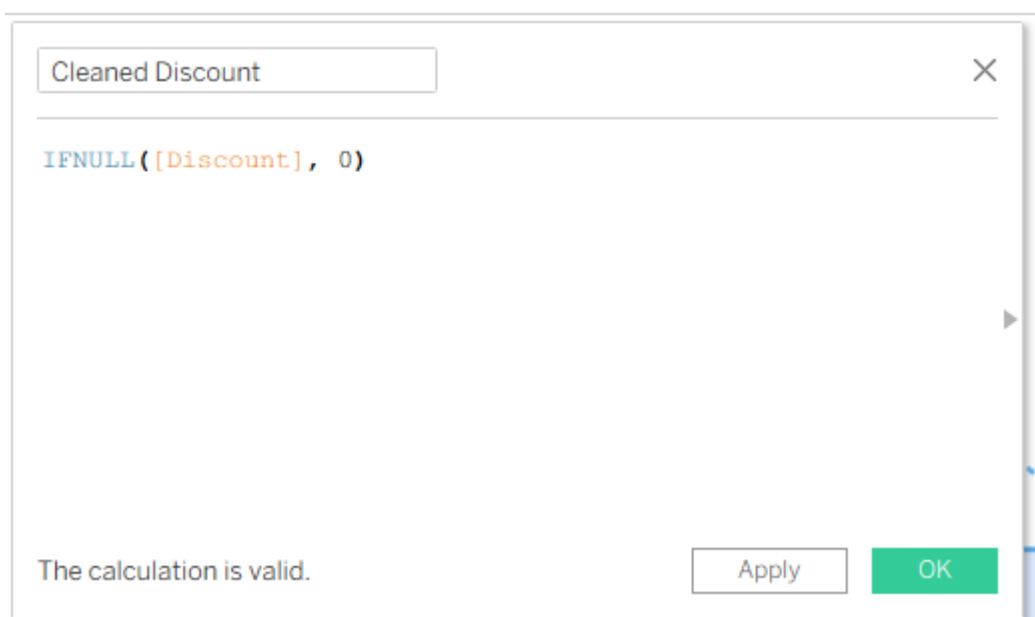
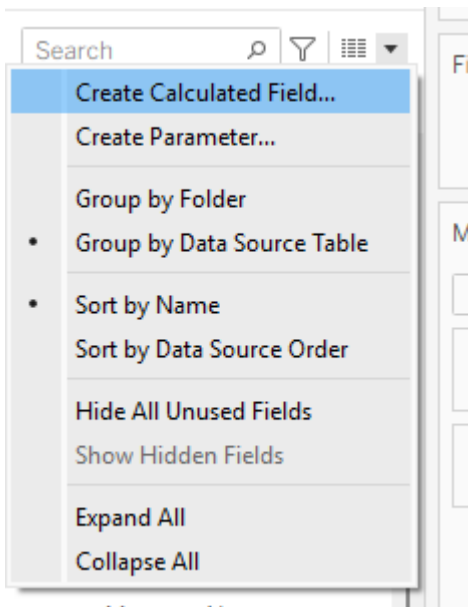
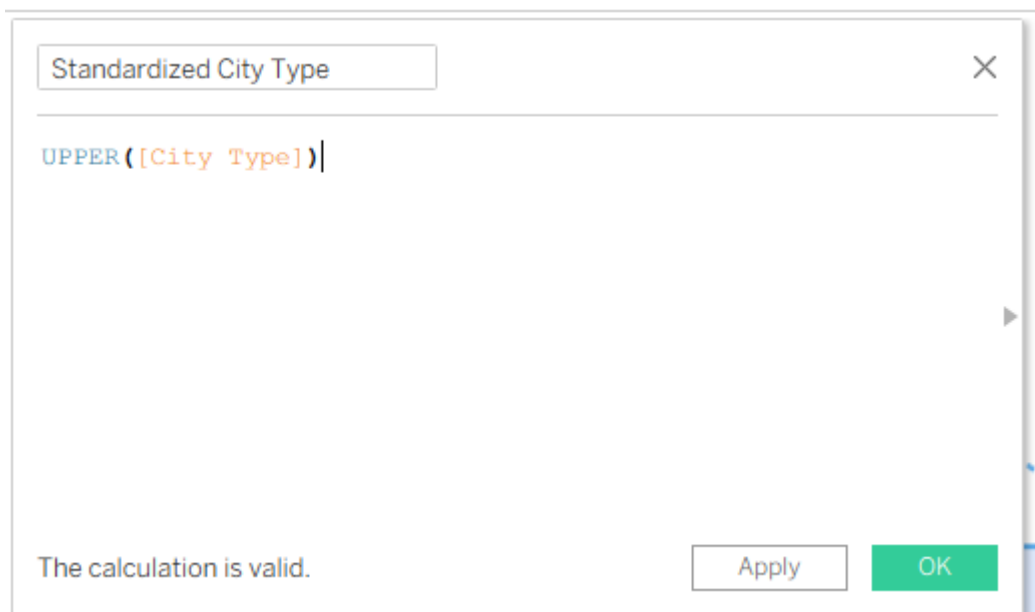




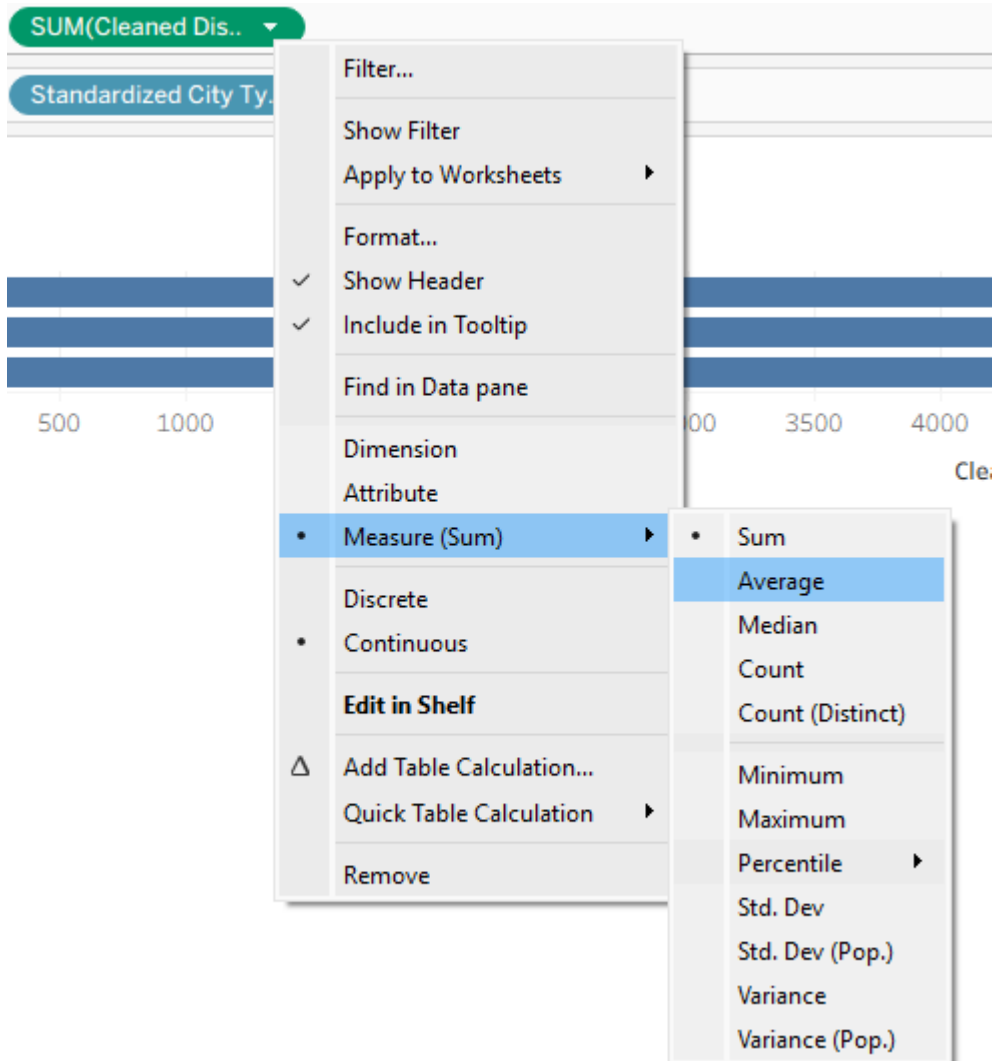
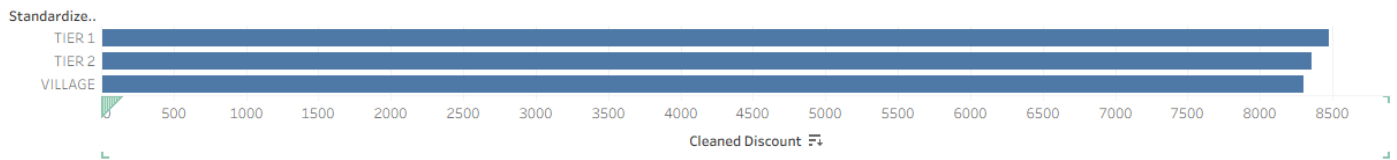
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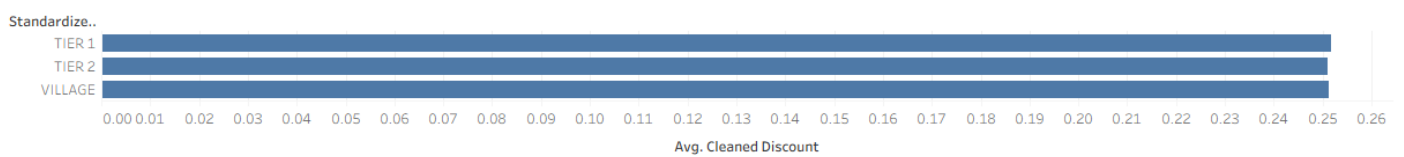
g) Apply calculations to fix data issues and enhance insights

**UPPER:**

Sheet 7



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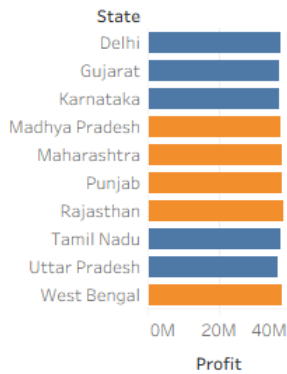
DASHBOARD :

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Sub-Catego..	Category of..
Apples	Sessional F..
Beds	Furniture
Buckets	Household I..
Burgers	Fast Food
Butter	Dairy Produ..
Carrots	Sessional F..
Chairs	Furniture
Cheese	Dairy Produ..
Detergents	Household I..
Fans	Electric Ap..
Fries	Fast Food
Mangoes	Sessional F..
Microwaves	Electric Ap..

OM
Ex..

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Product Na..
Apples - 119
Apples - 120
Apples - 121
Apples - 122
Apples - 123
Apples - 124
Apples - 125
Apples - 126
Apples - 127
Apples - 128
Apples - 129
Apples - 131
Apples - 132
Apples - 133
Apples - 134
Apples - 135
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Apples - 143
Apples - 144
Apples - 145
Apples - 146
Apples - 147
Apples - 148
Apples - 149
Apples - 150
Apples - 151

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Sub-Catego..
Apples
Beds
Buckets
Burgers
Butter
Carrots
Chairs
Cheese
Detergents
Fans
Fries
Mangoes
Microwaves
Milk
Mops
Pizzas
Refrigerato..
Sandwiches
Sofas
Tables
Tomatoes
Utensils
Washing M..
Yogurt

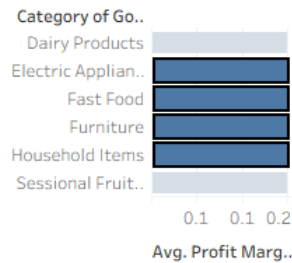
0.0 0.1 0.1 0.2
Aggregate Profit M..

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State	Sub-Catego..
Delhi	Apples
	Beds
	Buckets
	Burgers
	Butter
	Carrots
	Chairs
	Cheese
	Detergents
	Fans
	Fries
	Mangoes
	Microwaves

OM
Sal..

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KPI Threshold Met?

Below Target
Target Achieved

Select Profit Threshold

3,75,00,000

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Standardize..

TIER 1

TIER 2

VILLAGE

0.0 0.1

Avg. Clea..